Question 1

```
from sklearn.datasets import load_diabetes
import pandas as pd
import statsmodels.api as sm

diabetes = load_diabetes(as_frame=True)
df = diabetes.frame

print(df.head())
```

```
        age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6  target
0 -0.002592  0.019907 -0.017646   151.0
1 -0.039493 -0.068332 -0.092204    75.0
2 -0.002592  0.002861 -0.025930   141.0
3  0.034309  0.022688 -0.009362   206.0
4 -0.002592 -0.031988 -0.046641   135.0
```

```
# correlation
print(df[["bmi","target"]].corr())

# Simple linear regression
X = sm.add_constant(df["bmi"])
y = df['target']

model = sm.OLS(y,X).fit()

print(model.summary())
```

```
            bmi   target
bmi     1.00000  0.58645
target  0.58645  1.00000
                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.344
Model:                            OLS   Adj. R-squared:                  0.342
Method:                 Least Squares   F-statistic:                     230.7
Date:                Thu, 08 Oct 2026   Prob (F-statistic):           3.47e-42
Time:                        08:04:33   Log-Likelihood:                -2454.0
No. Observations:                 442   AIC:                             4912.
Df Residuals:                     440   BIC:                             4920.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        152.1335      2.974     51.162      0.000     146.289     157.978
bmi          949.4353     62.515     15.187      0.000     826.570    1072.301
==============================================================================
Omnibus:                       11.674   Durbin-Watson:                   1.848
Prob(Omnibus):                  0.003   Jarque-Bera (JB):                7.310
Skew:                           0.156   Prob(JB):                       0.0259
Kurtosis:                       2.453   Cond. No.                         21.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Interpretation : The entire workflow loads the scikit-learn Diabetes dataset into a pandas DataFrame, evaluates the linear correlation between Body Mass Index (bmi) and disease progression (target), and fits an Ordinary Least Squares (OLS) linear regression model to quantify how strongly bmi predicts target.

Question 2

```
X = df[["bmi","bp","age"]]
X = sm.add_constant(X)

model = sm.OLS(df["target"],X).fit()

print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.396
Model:                            OLS   Adj. R-squared:                  0.392
Method:                 Least Squares   F-statistic:                     95.81
Date:                Thu, 08 Oct 2026   Prob (F-statistic):           1.09e-47
Time:                        08:07:02   Log-Likelihood:                -2435.7
No. Observations:                 442   AIC:                             4879.
Df Residuals:                     438   BIC:                             4896.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        152.1335      2.859     53.211      0.000     146.514     157.753
bmi          788.7827     65.562     12.031      0.000     659.927     917.638
bp           394.1271     68.392      5.763      0.000     259.710     528.544
age           25.9904     63.923      0.407      0.685     -99.643     151.623
==============================================================================
Omnibus:                       12.586   Durbin-Watson:                   1.928
Prob(Omnibus):                  0.002   Jarque-Bera (JB):                7.934
Skew:                           0.175   Prob(JB):                       0.0189
Kurtosis:                       2.444   Cond. No.                         28.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

interpretation : Q2) The code performs a Multiple Linear Regression analysis using statsmodels to evaluate how three feature variables (bmi, bp, and age) jointly predict the disease progression outcome variable (target). It adds an intercept constant to the predictors, fits an Ordinary Least Squares (OLS) model, and outputs a detailed summary showing that the model accounts for roughly 39.6% of the variance ($R^2 = 0.396$), with bmi and bp serving as statistically significant predictors ($p < 0.001$) while age is not ($p = 0.685$).

Question 3

```
from scipy.stats import spearmanr

study_time = [2,4,1,5,3]
exam_score = [50,65,45,80,60]

correlation,p_value =spearmanr(study_time,exam_score)

print("Spearman Correlation:",correlation)
print("P-value:",p_value)
```

```
Spearman Correlation: 0.9999999999999999
P-value: 1.4042654220543672e-24
```

Interpretation : The code imports spearmanr from scipy.stats to calculate the Spearman rank correlation coefficient and p-value between two datasets, study_time and exam_score. The output demonstrates a perfect positive monotonic relationship ($\rho \approx 1.0$) with statistical significance ($p \approx 1.40 \times 10^{-24}$), indicating that as study time increases, exam scores consistently increase in rank order.